

Pandas vs Polars vs DuckDB

 **EuropeSales.csv**

[Enlace descarga](#)

```
pip install polars
```

```
import pandas as pd
import polars as pl
import duckdb
```

Carga de datos (CSVs)

Pandas

```
df_pd = pd.read_csv("EuropeSales.csv")
df_pd.head(3)
```

	Region	Country	Item Type	Sales Channel	Order Priority	Order Date	Order ID	Ship Date	Units Sold	Unit Price	Unit Cost	Total Revenue	Total Cost	Total Profit
0	Europe	Czech Republic	Beverages	Offline	C	9/12/2011	478051030	9/29/2011	4778	47.45	31.79	226716.10	151892.62	74823.48
1	Europe	Bosnia and Herzegovina	Clothes	Online	M	10/14/2013	919133651	11/4/2013	927	109.28	35.84	101302.56	33223.68	68078.88
2	Europe	Austria	Cereal	Offline	C	8/13/2014	987410676	9/6/2014	5616	205.70	117.11	1155211.20	657689.76	497521.44

Polars

```
df_pl = pl.read_csv("EuropeSales.csv")
df_pl.head(3)
```

	Region	Country	Item Type	Sales Channel	Order Priority	Order Date	Order ID	Ship Date	Units Sold	Unit Price	Unit Cost	Total Revenue	Total Cost	Total Profit
	str	str	str	str	str	str	i64	str	i64	f64	f64	f64	f64	f64
"Europe"	"Europe"	"Czech Republic"	"Beverages"	"Offline"	"C"	"9/12/2011"	478051030	"9/29/2011"	4778	47.45	31.79	226716.1	151892.62	74823.48
"Europe"	"Europe"	"Bosnia and Herzegovina"	"Clothes"	"Online"	"M"	"10/14/2013"	919133651	"11/4/2013"	927	109.28	35.84	101302.56	33223.68	68078.88
"Europe"	"Europe"	"Austria"	"Cereal"	"Offline"	"C"	"8/13/2014"	987410676	"9/6/2014"	5616	205.7	117.11	1155211.2	657689.76	497521.44

DuckDB

```
con = duckdb.connect()
df_duck = con.execute("SELECT * FROM 'EuropeSales.csv']").df()
df_duck.head(3)
```

	Region	Country	Item Type	Sales Channel	Order Priority	Order Date	Order ID	Ship Date	Units Sold	Unit Price	Unit Cost	Total Revenue	Total Cost	Total Profit
0	Europe	Czech Republic	Beverages	Offline	C	2011-09-12	478051030	2011-09-29	4778	47.45	31.79	226716.10	151892.62	74823.48
1	Europe	Bosnia and Herzegovina	Clothes	Online	M	2013-10-14	919133651	2013-11-04	927	109.28	35.84	101302.56	33223.68	68078.88
2	Europe	Austria	Cereal	Offline	C	2014-08-13	987410676	2014-09-06	5616	205.70	117.11	1155211.20	657689.76	497521.44

Filtrado de Datos

Pandas

```
fpd = df_pd[df_pd.Country == "Ukraine"]
fpd.head(10)
```

	Region	Country	Item Type	Sales Channel	Order Priority	Order Date	Order ID	Ship Date	Units Sold	Unit Price	Unit Cost	Total Revenue	Total Cost	Total Profit
15	Europe	Ukraine	Cosmetics	Online	H	1/22/2011	773645913	1/28/2011	7873	437.20	263.33	3442075.60	2073197.09	1368878.51
25	Europe	Ukraine	Personal Care	Online	M	7/16/2010	987459170	8/27/2010	9967	81.73	56.67	814602.91	564829.89	249773.02
156	Europe	Ukraine	Cosmetics	Online	C	6/9/2014	183910119	6/11/2014	1946	437.20	263.33	850791.20	512440.18	338351.02
164	Europe	Ukraine	Snacks	Offline	C	1/25/2016	611791121	3/3/2016	1443	152.58	97.44	220172.94	140605.92	79567.02
226	Europe	Ukraine	Office Supplies	Offline	M	5/3/2017	181260526	6/3/2017	7872	651.21	524.96	5126325.12	4132485.12	993840.00
292	Europe	Ukraine	Vegetables	Online	C	11/6/2011	446019017	11/25/2011	5794	154.06	90.93	892623.64	526848.42	365775.22
296	Europe	Ukraine	Clothes	Offline	H	9/27/2016	747955071	10/25/2016	5382	109.28	35.84	588144.96	192890.88	395254.08
343	Europe	Ukraine	Household	Online	H	7/28/2010	741866242	9/13/2010	3562	668.27	502.54	2380377.74	1790047.48	590330.26
407	Europe	Ukraine	Meat	Offline	H	4/25/2017	783594977	5/1/2017	2022	421.89	364.69	853061.58	737403.18	115658.40
488	Europe	Ukraine	Cosmetics	Online	C	7/8/2014	140461766	7/20/2014	4613	437.20	263.33	2016803.60	1214741.29	802062.31

Polars

```
fpd = df_pl.filter(pl.col("Country") == "Ukraine")
fpd.head(10)
```

	Region	Country	Item Type	Sales Channel	Order Priority	Order Date	Order ID	Ship Date	Units Sold	Unit Price	Unit Cost	Total Revenue	Total Cost	Total Profit
	str	str	str	str	str	str	i64	str	i64	f64	f64	f64	f64	f64
	"Europe"	"Ukraine"	"Cosmetics"	"Online"	"H"	"1/22/2011"	773645913	"1/28/2011"	7873	437.2	263.33	3442075.6	2.0732e6	1.3689e6
	"Europe"	"Ukraine"	"Personal Care"	"Online"	"M"	"7/16/2010"	987459170	"8/27/2010"	9967	81.73	56.67	814602.91	564829.89	249773.02
	"Europe"	"Ukraine"	"Cosmetics"	"Online"	"C"	"6/9/2014"	183910119	"6/11/2014"	1946	437.2	263.33	850791.2	512440.18	338351.02
	"Europe"	"Ukraine"	"Snacks"	"Offline"	"C"	"1/25/2016"	611791121	"3/3/2016"	1443	152.58	97.44	220172.94	140605.92	79567.02
	"Europe"	"Ukraine"	"Office Supplies"	"Offline"	"M"	"5/3/2017"	181260526	"6/3/2017"	7872	651.21	524.96	5.1263e6	4.1325e6	993840.0
	"Europe"	"Ukraine"	"Vegetables"	"Online"	"C"	"11/6/2011"	446019017	"11/25/2011"	5794	154.06	90.93	892623.64	526848.42	365775.22
	"Europe"	"Ukraine"	"Clothes"	"Offline"	"H"	"9/27/2016"	747955071	"10/25/2016"	5382	109.28	35.84	588144.96	192890.88	395254.08
	"Europe"	"Ukraine"	"Household"	"Online"	"H"	"7/28/2010"	741866242	"9/13/2010"	3562	668.27	502.54	2.3804e6	1.7900e6	590330.26
	"Europe"	"Ukraine"	"Meat"	"Offline"	"H"	"4/25/2017"	783594977	"5/1/2017"	2022	421.89	364.69	853061.58	737403.18	115658.4
	"Europe"	"Ukraine"	"Cosmetics"	"Online"	"C"	"7/8/2014"	140461766	"7/20/2014"	4613	437.2	263.33	2016803.6	1.2147e6	802062.31

DuckDB

```
df_duck = con.execute("SELECT * FROM 'EuropeSales.csv' WHERE Country LIKE 'Ukraine'").df()
df_duck.head(10)
```

	Region	Country	Item Type	Sales Channel	Order Priority	Order Date	Order ID	Ship Date	Units Sold	Unit Price	Unit Cost	Total Revenue	Total Cost	Total Profit
0	Europe	Ukraine	Cosmetics	Online	H	2011-01-22	773645913	2011-01-28	7873	437.20	263.33	3442075.60	2073197.09	1368878.51
1	Europe	Ukraine	Personal Care	Online	M	2010-07-16	987459170	2010-08-27	9967	81.73	56.67	814602.91	564829.89	249773.02
2	Europe	Ukraine	Cosmetics	Online	C	2014-06-09	183910119	2014-06-11	1946	437.20	263.33	850791.20	512440.18	338351.02
3	Europe	Ukraine	Snacks	Offline	C	2016-01-25	611791121	2016-03-03	1443	152.58	97.44	220172.94	140605.92	79567.02

	Region	Country	Item Type	Sales Channel	Order Priority	Order Date	Order ID	Ship Date	Units Sold	Unit Price	Unit Cost	Total Revenue	Total Cost	Total Profit
4	Europe	Ukraine	Office Supplies	Offline	M	2017-05-03	181260526	2017-06-03	7872	651.21	524.96	5126325.12	4132485.12	993840.00
5	Europe	Ukraine	Vegetables	Online	C	2011-11-06	446019017	2011-11-25	5794	154.06	90.93	892623.64	526848.42	365775.22
6	Europe	Ukraine	Clothes	Offline	H	2016-09-27	747955071	2016-10-25	5382	109.28	35.84	588144.96	192890.88	395254.08
7	Europe	Ukraine	Household	Online	H	2010-07-28	741866242	2010-09-13	3562	668.27	502.54	2380377.74	1790047.48	590330.26
8	Europe	Ukraine	Meat	Offline	H	2017-04-25	783594977	2017-05-01	2022	421.89	364.69	853061.58	737403.18	115658.40
9	Europe	Ukraine	Cosmetics	Online	C	2014-07-08	140461766	2014-07-20	4613	437.20	263.33	2016803.60	1214741.29	802062.31

Agregar Datos

Pandas

```
agg_pd = df_pd.groupby("Country")["Total Revenue"].sum().reset_index()
agg_pd.head(10)
```

	Country	Total Revenue
0	Albania	32224853.87
1	Andorra	47756693.17
2	Armenia	37519840.21
3	Austria	35740871.49
4	Belarus	34236260.77
5	Belgium	25852572.30
6	Bosnia and Herzegovina	50117508.49
7	Bulgaria	38161555.70
8	Croatia	27348195.70
9	Cyprus	33008851.50

Polars

Aggregation

Aggregation Expressions

```
agg_pl = df_pl.group_by("Country").agg([pl.sum("Total Revenue").alias("country_revenue")])
agg_pl.head(15)
```

Country	country_revenue
str	f64
"Switzerland"	3.1875174e7
"Liechtenstein"	2.9873e7
"Italy"	3.5878e7
"Croatia"	2.7348e7
"Slovakia"	4.2941e7
...	...
"Armenia"	3.7520e7
"Macedonia"	4.9222e7
"Montenegro"	3.1346e7
"Latvia"	3.8722e7
"Austria"	3.5741e7

DuckDB

```
agg_duck = con.execute("""
    SELECT Country, sum("Total Revenue") AS country_revenue
    FROM 'EuropeSales.csv'
    GROUP BY country
    ORDER BY country
""").df()
agg_duck.head(10)
```

	Country	country_revenue
0	Albania	32224853.87
1	Andorra	47756693.17
2	Armenia	37519840.21
3	Austria	35740871.49
4	Belarus	34236260.77
5	Belgium	25852572.30
6	Bosnia and Herzegovina	50117508.49
7	Bulgaria	38161555.70
8	Croatia	27348195.70
9	Cyprus	33008851.50